

SUPPLY CHAIN DELIVERY PERFORMANCE & DELAY RISK ANALYSIS

An end-to-end analytics project covering raw-data preparation, KPI design, EDA, business diagnosis, visualization, and predictive-model workflow.

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Target Role: Data Analyst

Technology: Python-based Analytics & Machine Learning

Tools: Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, imbalanced-learn

Project Type: Supply Chain / Operational Analytics

Date: 05 October 2026

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1. Executive Summary

This project analyzes delivery performance and profitability in a large supply-chain dataset. The raw file contains 180,519 records and 53 columns. After the cleaning logic in the supplied Python script, the working dataset contains 172,765 records and 29 retained columns.

The completed analytical layer identifies 94,523 positively delayed orders (54.71%) and an aggregate Order Profit Per Order of \$3,806,421. The analysis examines operational bottlenecks, profitability, regional patterns, time patterns, and a predictive late-delivery modelling workflow.

2. Business Problem

The business problem is to understand where delivery reliability is breaking down, quantify the scale of delay, identify the operational dimensions associated with higher delay rates, and translate those findings into practical improvement priorities.

Business Questions -

- How large is the delivery-delay problem after cleaning?
- Which regions, customer segments, shipping modes, statuses and departments show higher delay rates?
- How does delay relate to order-level profitability?
- Are there meaningful monthly, weekday or hourly patterns?
- Which variables can support proactive late-delivery prediction?

3. Project Objectives

1. Prepare a reliable analytical dataset from the raw supply-chain file.
2. Measure delivery performance through delay-based KPIs.
3. Identify operational bottlenecks across business dimensions.
4. Assess the relationship between delay and profitability.
5. Analyze regional and time-based delivery patterns.
6. Investigate regional delay drivers in Central Africa and East Africa.
7. Implement and compare classification workflows for late-delivery risk.
8. Translate evidence into practical business recommendations.

4. Dataset Overview

Attribute	Details
Dataset	DataCoSupplyChainDataset.csv
Raw size	180,519 records × 53 columns
Analytical size	172,765 records × 29 columns
Order date range	01 Jan 2015 – 31 Jan 2018
Key dimensions	Order Region, Customer Segment, Shipping Mode, Order Status, Department Name, Type
Key measures	Sales, Sales per customer, Order Profit Per Order, Order Item Quantity, shipping/processing days
Raw duplicate rows	0
Working missing values	No missing values in retained working columns after date validation

The raw file contains order, customer, product, shipment, financial, geographic and operational fields. The script removes customer-identifying fields and other fields that are not required for this analysis.

5. Technology Stack

Technology	Purpose
Python	End-to-end analysis and modelling orchestration
Pandas	Data ingestion, cleaning, grouping, transformation and KPIs
NumPy	Numerical operations and derived profitability logic
Matplotlib	Business charts
Seaborn	Statistical/business visualization
scikit-learn	Preprocessing, Logistic Regression, Random Forest, metrics and validation
imbalanced-learn	SMOTE for the balanced Random Forest workflow

6. Data Preparation & ETL

Data Loading -

The script reads the CSV with latin1 encoding and inspects columns, dimensions, duplicates and missing values.

Data Cleaning -

It removes redundant IDs, descriptive/image fields, customer-identifying fields, coordinates, discount-detail fields and other fields outside the analytical focus.

Cancelled Orders -

Rows with Delivery Status = Shipping canceled are excluded.

Date Handling -

Order and shipping timestamps are converted to datetime; invalid dates are removed.

Duplicate Handling -

Duplicate rows are checked and removed.

Feature Engineering -

Order Processing Time, Delay, Is_Delayed, order month, weekday, order hour and Profitability Flag are derived.

Validation -

Shape, missingness, data types, final columns and low-cardinality distributions are checked after cleaning.

Workflow: Raw CSV → Quality Checks → Field Selection → Cancelled-Order Removal → Date Validation → Feature Engineering → KPI/EDA → Predictive Modelling.

7. KPI Framework

KPI	Validated value	Business meaning
Total analytical orders	172,765	Volume covered by the cleaned analysis
Late deliveries	94,523	Orders with Delay > 0
Late delivery rate	54.71%	Core delivery-reliability indicator
90th percentile positive delay	3 days	Upper-end delay severity
Aggregate Order Profit Per Order	\$3,806,421	Aggregate order-level profitability
Profit-positive orders	139,354	Orders with positive profit
Loss-making orders	32,295	Orders with negative profit

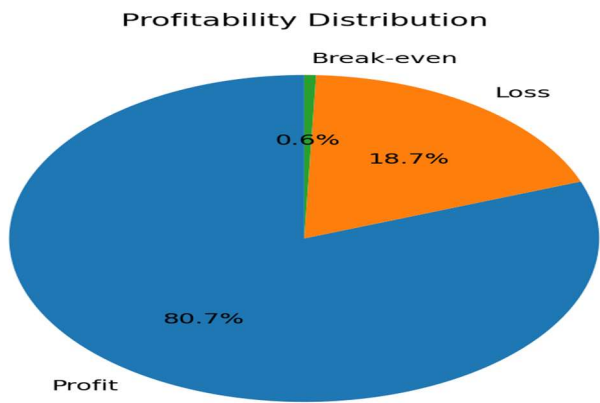
8. Exploratory Data Analysis

Delivery reliability is the dominant operational issue in the cleaned data: 54.71% of records have a positive derived delay. The 90th percentile of positive delay is 3 days.

Profitability is mixed: 80.66% of orders are profit-positive, 18.69% are loss-making and 0.65% are break-even.

9. Visual Analysis

Figure 1: Profitability Distribution



Key insight: Profit-positive orders represent 80.66% of the cleaned dataset.

Business interpretation: The operation is profitable at order level overall, but the loss-making population is material.

Recommendation: Segment loss-making orders by delay, region and operational attributes before targeting corrective actions.

Figure 2: Order Delay Distribution

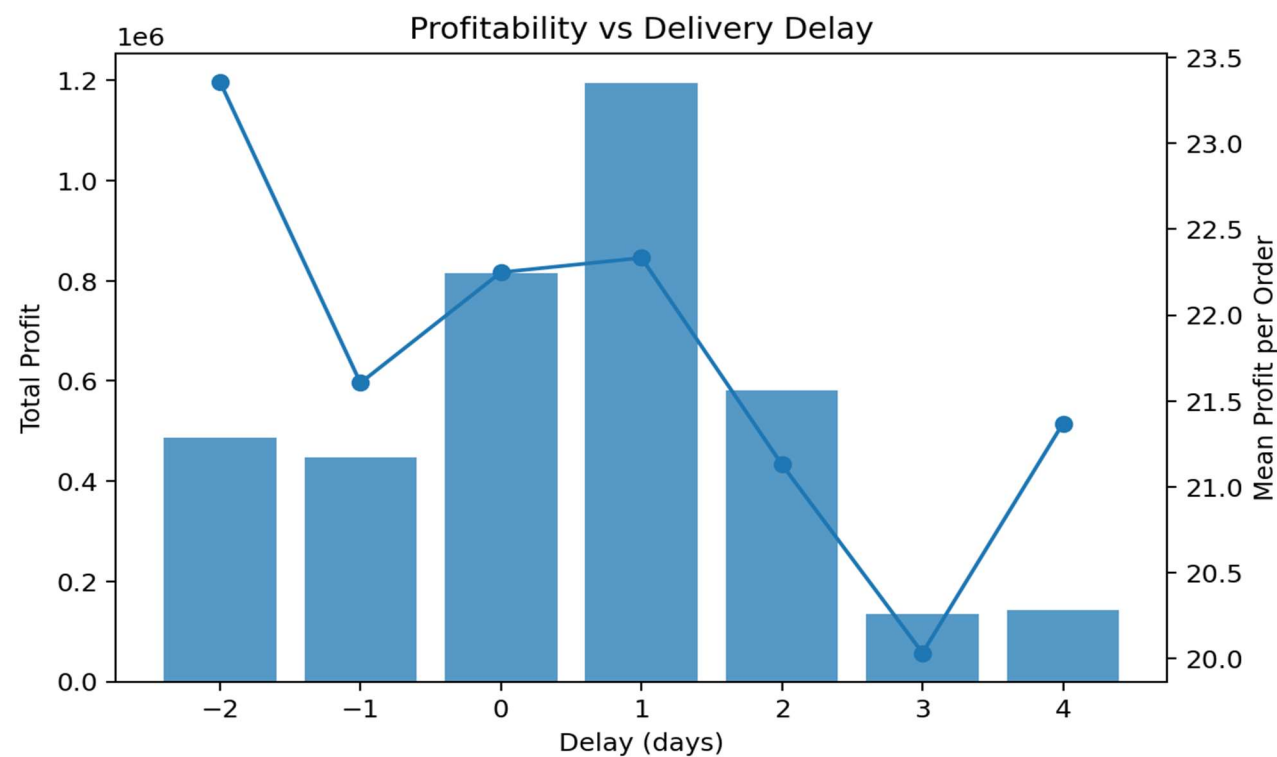


Key insight: One-day positive delays are the largest delay bucket at 30.97% of analytical orders.

Business interpretation: The delay problem is concentrated in relatively short delays.

Recommendation: Prioritize preventing the first day of delay rather than focusing only on extreme delays.

Figure 3: Profitability vs Delivery Delay

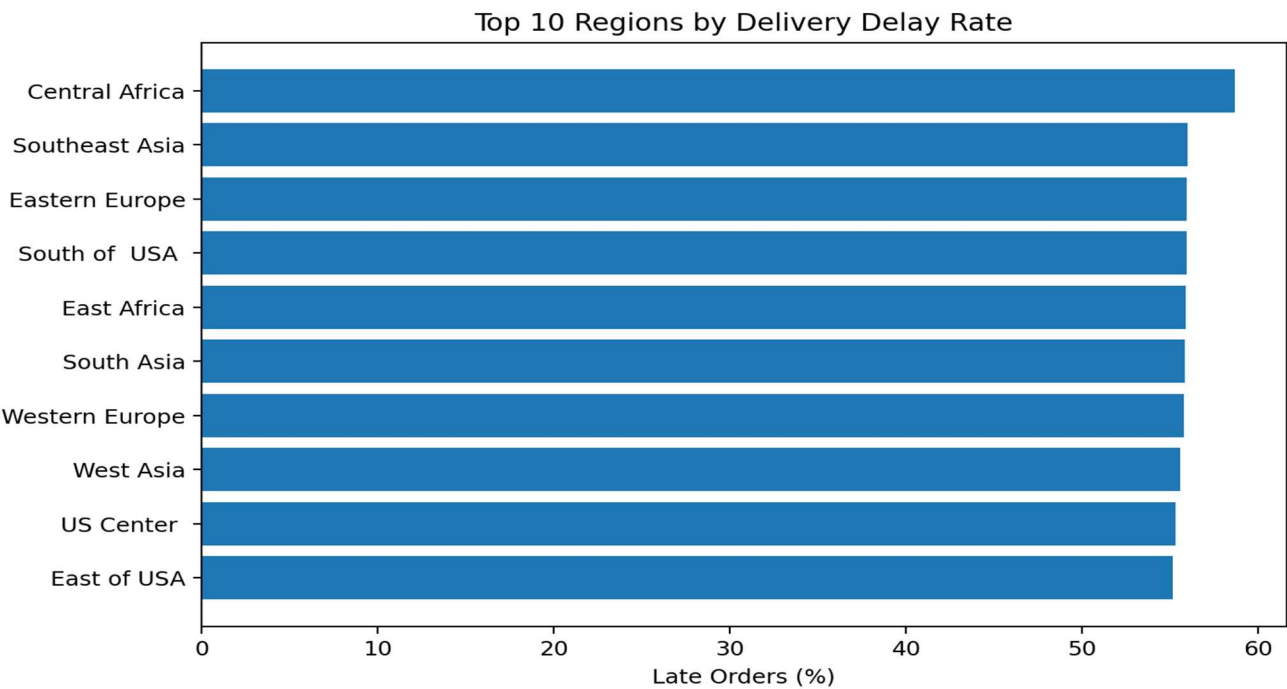


Key insight: The one-day-delay bucket contributes the largest total profit in the completed analysis; mean profit falls at three days.

Business interpretation: Delay and profitability are not a simple one-directional relationship.

Recommendation: Monitor delay and profitability together instead of assuming every delayed order is loss-making.

Figure 4: Top 10 Regions by Delivery Delay Rate

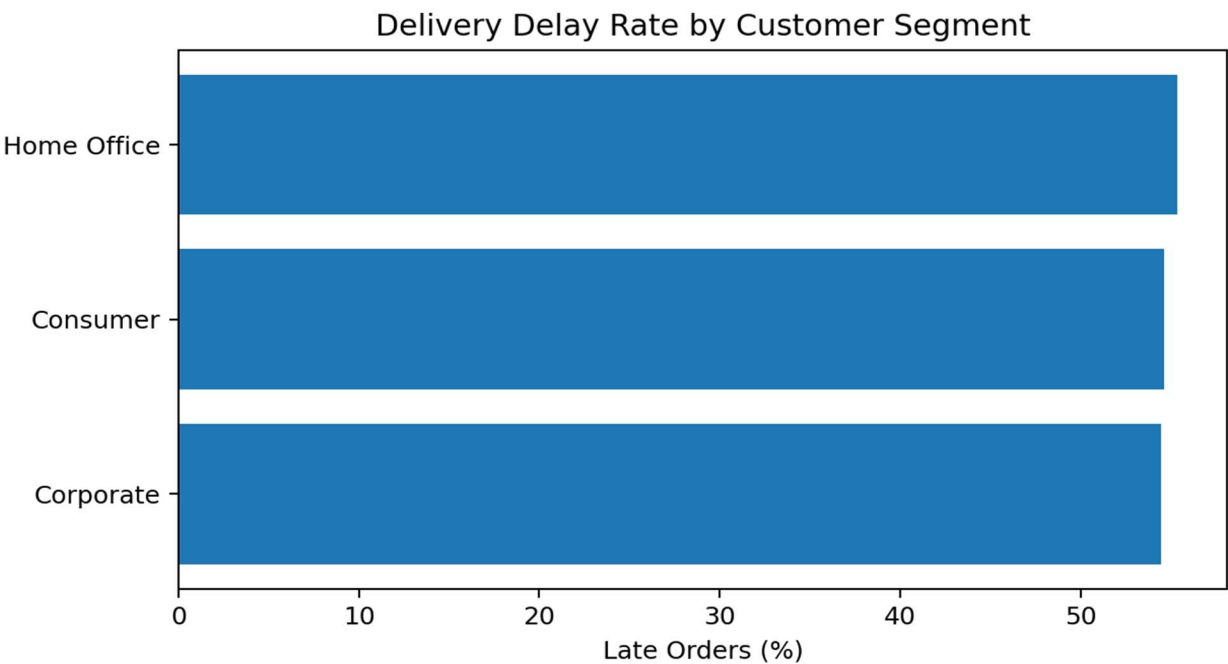


Key insight: Central Africa has the highest regional delay rate at 58.66%.

Business interpretation: Regional concentration provides a practical starting point for operational review.

Recommendation: Investigate carrier capacity, routing and service-level execution in high-delay regions.

Figure 5: Delivery Delay by Customer Segment

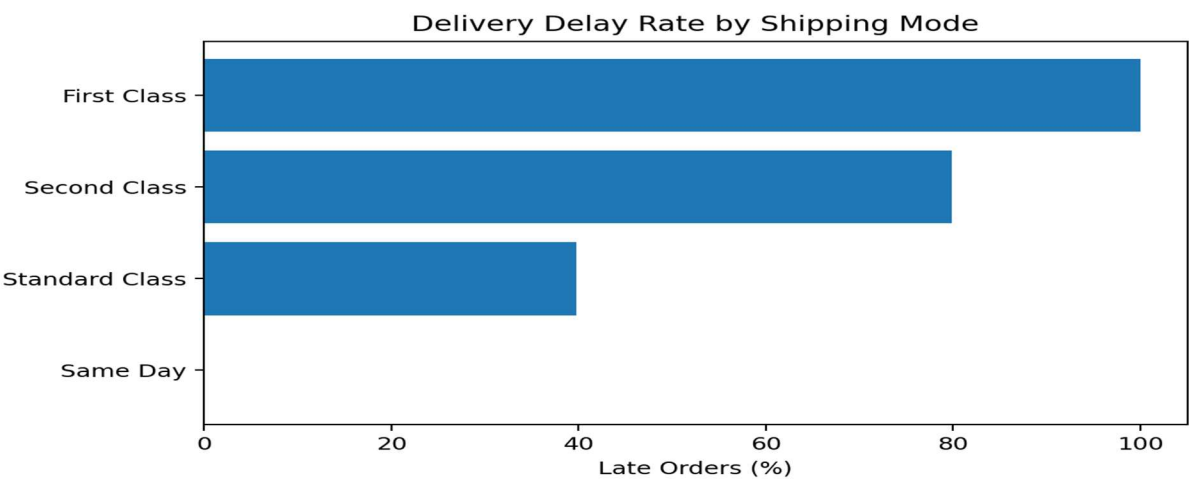


Key insight: Home Office has the highest segment delay rate at 55.36%.

Business interpretation: Segment differences are present but comparatively narrow.

Recommendation: Use segment as a supporting dimension rather than treating it as the primary root cause.

Figure 6: Delivery Delay by Shipping Mode

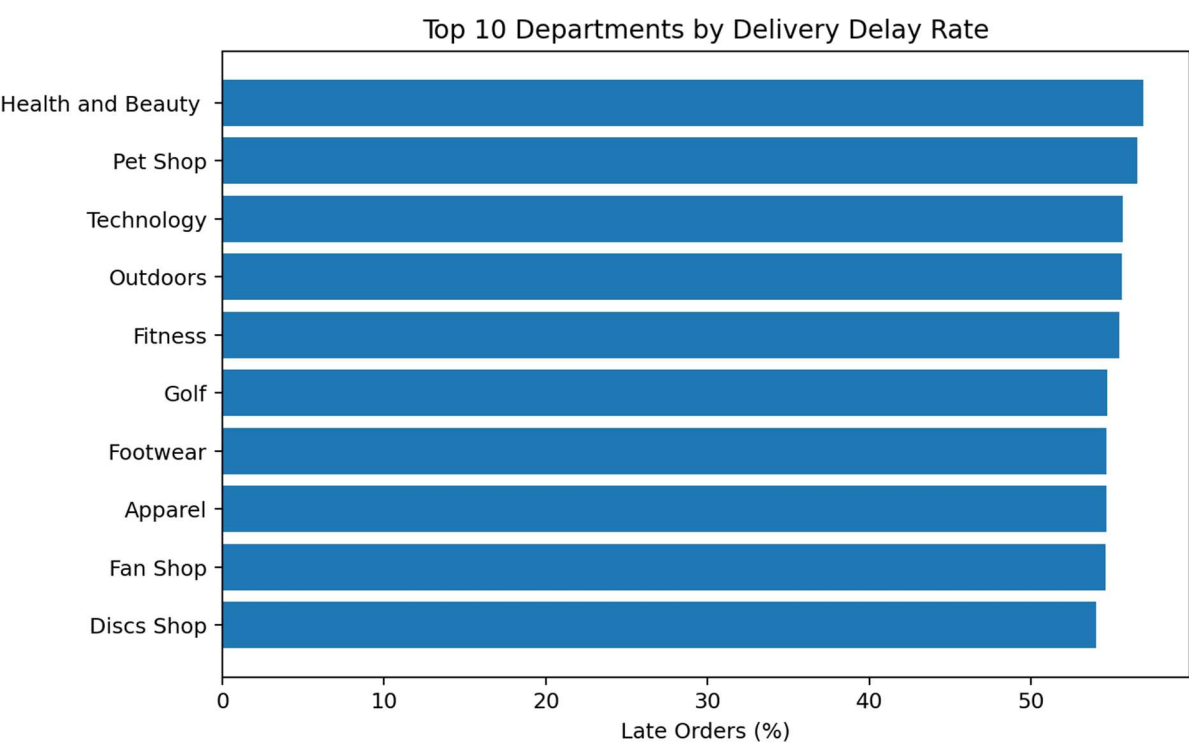


Key insight: First Class = 100.00%, Second Class = 79.83%, Standard Class = 39.77%, Same Day = 0.00% positive delay under the project definition.

Business interpretation: Shipping mode is one of the clearest operational differentiators.

Recommendation: Validate service-level definitions and timestamp rules before changing policy.

Figure 7: Top 10 Departments by Delivery Delay Rate

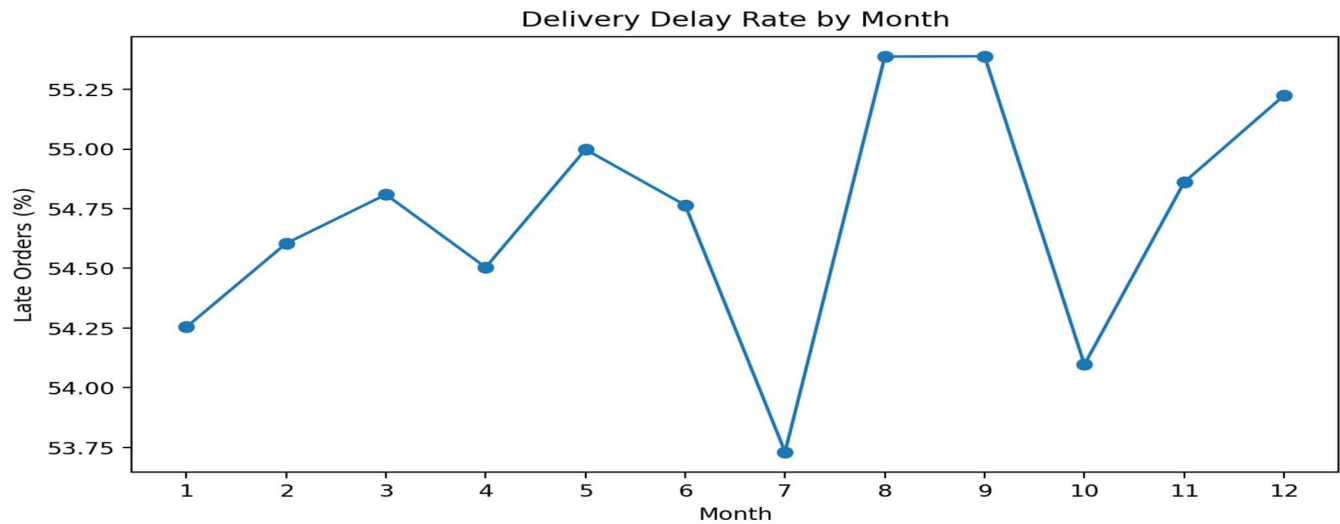


Key insight: Health and Beauty has the highest displayed department delay rate at 56.94%.

Business interpretation: Department differences are smaller than the shipping-mode gap.

Recommendation: Use department jointly with stronger operational variables such as region and shipping mode.

Figure 8: Delivery Delay by Month

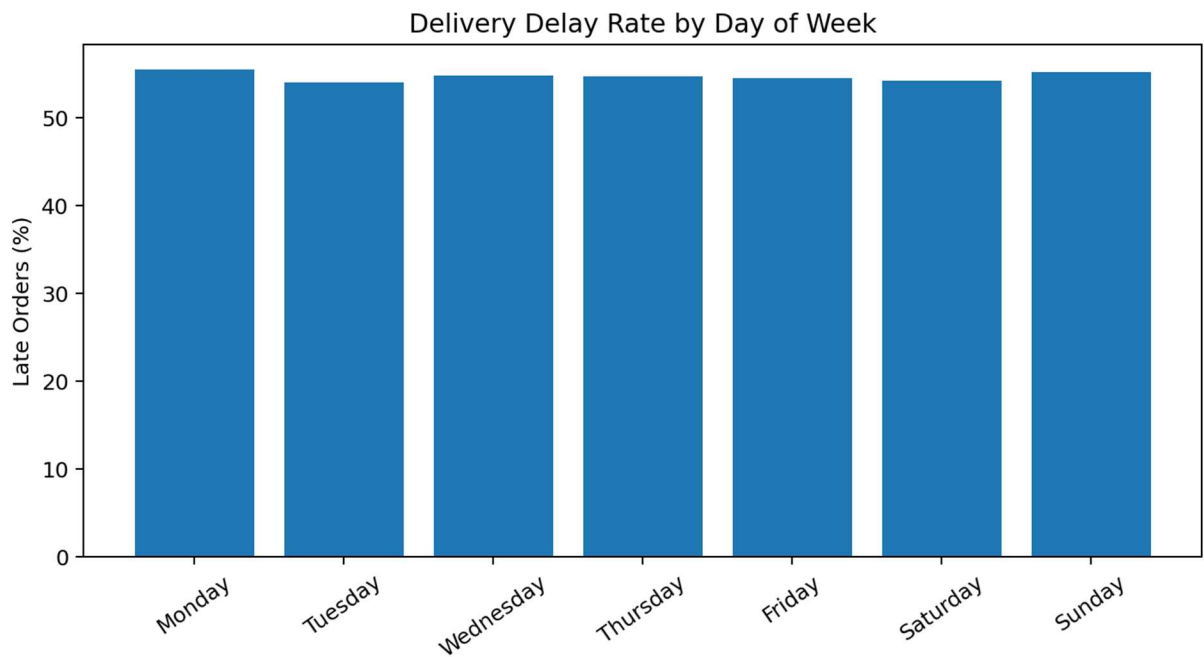


Key insight: September and August are the highest months at about 55.39%; July is lowest at about 53.73%.

Business interpretation: Monthly variation is visible but modest.

Recommendation: Use month for monitoring and planning, not as a standalone root cause.

Figure 9: Delivery Delay by Day of Week

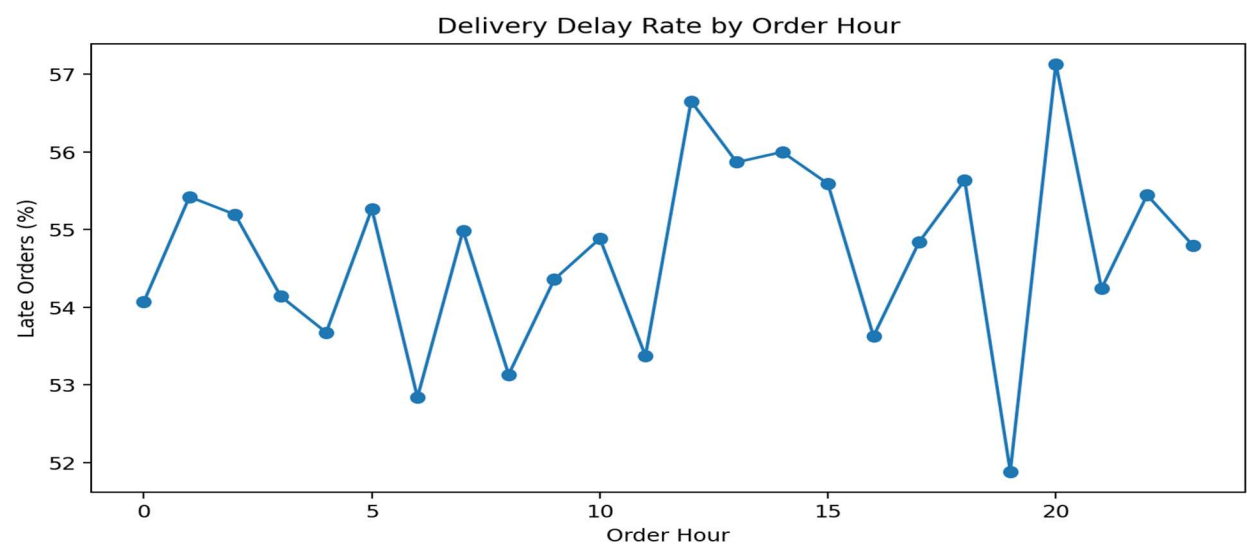


Key insight: Monday is highest at 55.53%; Tuesday is lowest at 53.98%.

Business interpretation: Weekday variation is relatively small.

Recommendation: Use weekday patterns for staffing/monitoring rather than broad policy changes.

Figure 10: Delivery Delay by Order Hour

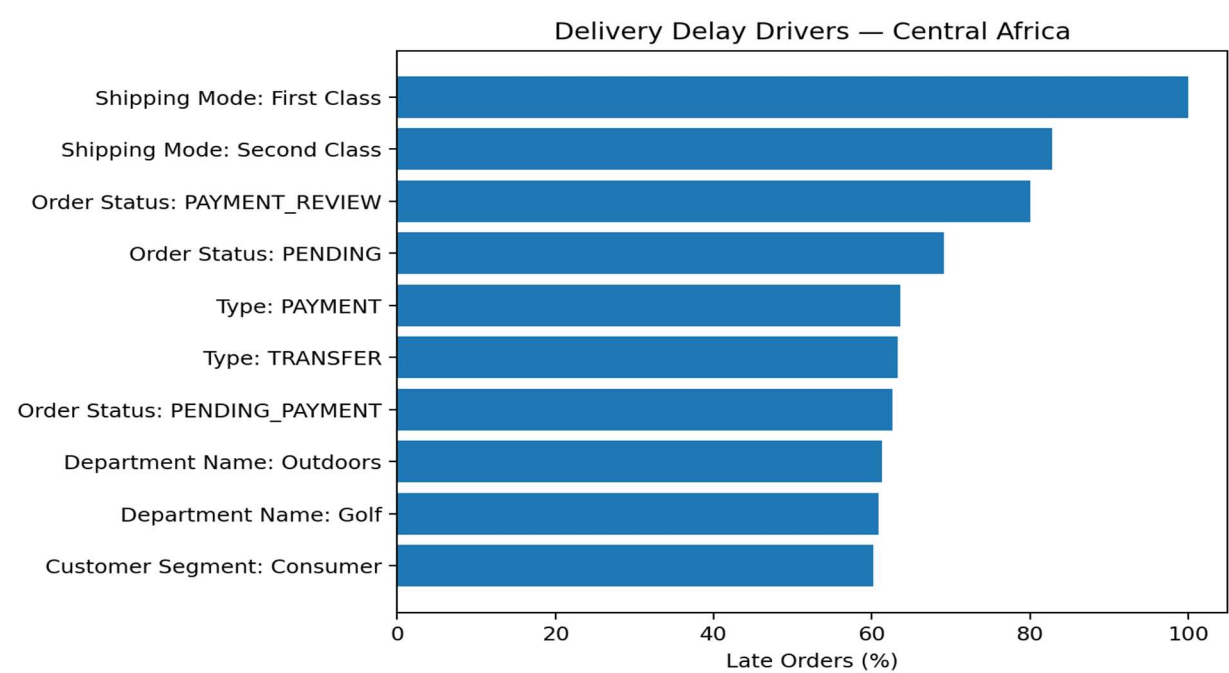


Key insight: 20:00 has the highest hourly rate at 57.13%; 19:00 is lowest at 51.88%.

Business interpretation: Hourly variation exists but is not dominant.

Recommendation: Combine hourly patterns with operational factors before taking action.

Figure 11: Delivery Delay Drivers — Central Africa

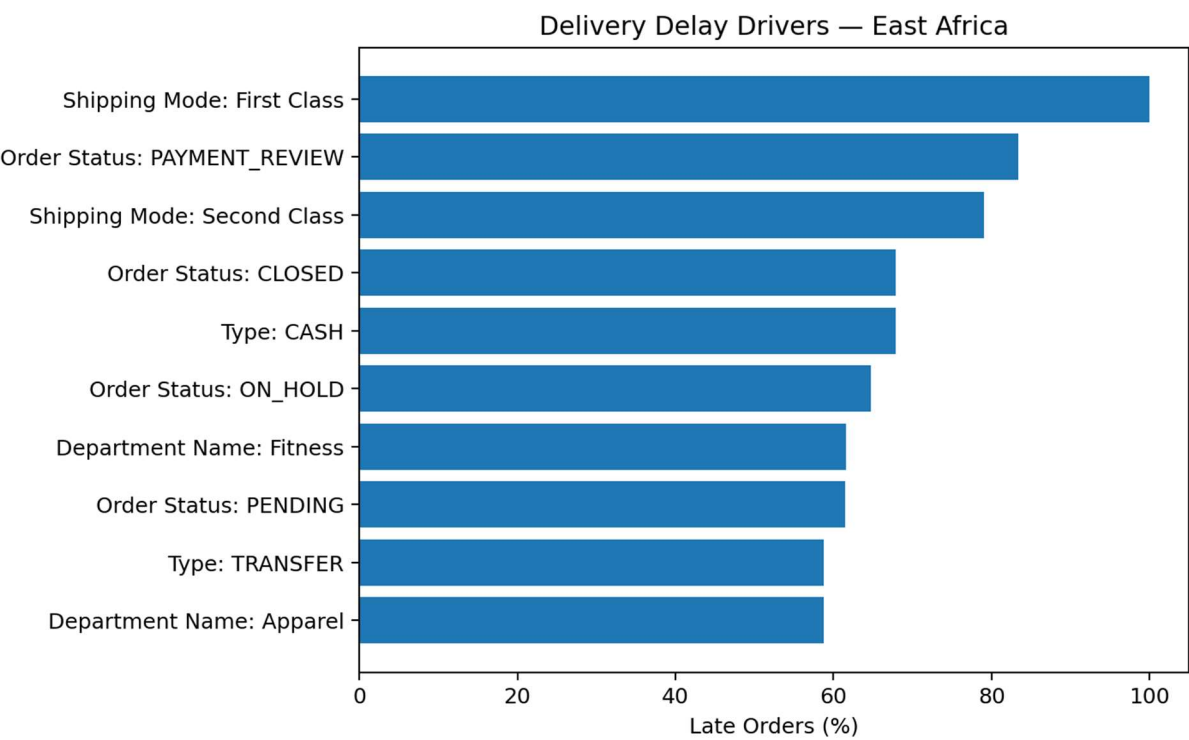


Key insight: The script ranks shipping mode, segment, department, type and status combinations by delay rate within Central Africa.

Business interpretation: This narrows a regional issue into specific factor levels for investigation.

Recommendation: Review high-delay factor levels together with order volume before changing regional operations.

Figure 12: Delivery Delay Drivers — East Africa



Key insight: The same factor-level screening is applied to East Africa.

Business interpretation: Comparing regions helps determine whether the same intervention should be applied consistently.

Recommendation: Treat driver rankings as diagnostic hypotheses rather than causal proof.

10. Graph Index

Figure	Visualization	Business purpose
Figure 1	Profitability Distribution	Answers an operational performance question
Figure 2	Order Delay Distribution	Answers an operational performance question
Figure 3	Profitability vs Delivery Delay	Answers an operational performance question
Figure 4	Top 10 Regions by Delivery Delay Rate	Answers an operational performance question
Figure 5	Delivery Delay by Customer Segment	Answers an operational performance question
Figure 6	Delivery Delay by Shipping Mode	Answers an operational performance question

Figure 7	Top 10 Departments by Delivery Delay Rate	Answers an operational performance question
Figure 8	Delivery Delay by Month	Answers an operational performance question
Figure 9	Delivery Delay by Day of Week	Answers an operational performance question
Figure 10	Delivery Delay by Order Hour	Answers an operational performance question
Figure 11	Delivery Delay Drivers — Central Africa	Answers an operational performance question
Figure 12	Delivery Delay Drivers — East Africa	Answers an operational performance question

11. Key Findings

- 94,523 of 172,765 cleaned records are positively delayed, producing a 54.71% late-delivery rate.
- Central Africa has the highest regional delay rate at 58.66%.
- Shipping mode shows the strongest visible separation: First Class and Second Class have much higher derived delay rates than Standard Class.
- Monthly, weekday and hourly patterns are comparatively narrow and should be treated as supporting signals.
- 32,295 orders are loss-making, so profitability should be monitored alongside delivery performance.

12. Business Recommendations

Priority	Problem / Evidence	Recommended action
High	First Class and Second Class show 100.00% and 79.83% delay rates.	Audit scheduled-day definitions, carrier/service mappings and shipment timestamps.
High	Central Africa has a 58.66% delay rate.	Run a regional driver review focused on high-volume, high-delay factor combinations.
High	One-day delays represent 30.97% of all analytical orders.	Introduce early-warning intervention before an order crosses its scheduled threshold.
Medium	32,295 orders are loss-making.	Combine profitability, delay and operational dimensions to identify commercially important problem segments.
Low	Time-based differences are relatively modest.	Use time patterns for monitoring and staffing rather than as primary root causes.

13. Areas Requiring Attention

Issue	Evidence	Interpretation	Action
High late-delivery rate	54.71% positive-delay rate	Indicates broad service-reliability pressure; cause is not proven.	Validate operational process and service-level configuration.
First Class anomaly	100% positive-delay rate	Could reflect the project's scheduled-vs-real calculation.	Audit source-system definitions before concluding there is universal failure.

Regional concentration	Central Africa 58.66%	Potential route/capacity/process differences; causal attribution is not established.	Investigate factor-level patterns with operations teams.
Loss-making population	32,295 loss orders	Profitability is affected by more than delay alone.	Analyze losses across product, region, segment and shipment attributes.

14. Technical Implementation

- Data Source → DataCoSupplyChainDataset.csv
- Data Ingestion → Pandas read_csv
- Data Cleaning → field reduction, cancelled-order removal, date conversion and duplicate/date validation
- Transformation → processing time, delay, delayed flag, calendar features and profitability flag
- EDA → descriptive statistics, bottleneck analysis, profitability/delay analysis and time-based analysis
- KPI Development → order volume, late deliveries, delay percentile, delivery rates and aggregate profit
- Visualization → Matplotlib and Seaborn
- Predictive Modelling → Logistic Regression, Random Forest and SMOTE-based Balanced Random Forest
- Validation → stratified train/test split, classification metrics, ROC-AUC, confusion matrices and 5-fold stratified cross-validation in the supplied code
- Risk Layer → delay probability, Low/Medium/High risk segmentation and prediction error analysis

15. Predictive Modelling

The supplied code implements a late-delivery classification workflow with Is_Delayed as the target. It explicitly removes outcome-derived variables such as Delay, Delivery Status, shipping date, Order Processing Time and Late_delivery_risk from the modelling features to reduce leakage. Categorical variables are one-hot encoded; the data is split 80/20 with stratification; and Logistic Regression, Random Forest and SMOTE-based Balanced Random Forest are defined.

Component	Implementation
Target	Is_Delayed
Train/test split	80/20, stratified, random_state=42
Categorical preprocessing	OneHotEncoder(handle_unknown='ignore')
Baseline	Logistic Regression, max_iter=1000
Tree model	Random Forest, 200 estimators
Balanced model	SMOTE + Random Forest, 200 estimators
Metrics in code	Accuracy, Precision, Recall, F1, ROC-AUC
Validation in code	5-fold StratifiedKFold with F1 scoring
Risk bands	0–0.30 Low, 0.30–0.70 Medium, 0.70–1.00 High

16. Dashboard / Reporting Layer

No Power BI, Excel, Tableau, DAX or Power Query implementation is present in the supplied project files. The implemented reporting layer is Python-based. A production dashboard could surface the validated KPIs, regional/shipping bottlenecks, profitability vs delay, time trends and validated predictive risk.

17. Challenges Faced

- Large raw dataset requiring deliberate field selection.
- Substantial raw missingness in fields such as Product Description and Order Zipcode.
- Date parsing and invalid-date validation.
- Avoiding target leakage in late-delivery prediction.
- Class-balancing considerations leading to the SMOTE workflow.

18. Limitations

- Historical data does not represent real-time operations.
- Delay-driver analysis identifies associations, not causal effects.
- Shipping-mode anomalies should be validated against business definitions.
- No financial savings/ROI estimate is calculated from the supplied analysis.
- Predictive model scores were not validated because the full model execution did not complete.
- No interactive BI dashboard was supplied.

19. Future Scope

- Run and tune the predictive models in a production-capable environment.
- Build an interactive Power BI reporting layer.
- Automate the data-to-report pipeline.
- Add carrier, warehouse, route, inventory and capacity data for deeper diagnosis.
- Monitor model drift and recalibrate risk thresholds.
- Track intervention outcomes to measure whether late-delivery rates improve.

20. Conclusion

This project demonstrates a practical supply-chain analytics workflow from raw data to business recommendations. The completed analysis processes 180K+ source records, creates delivery and profitability features, quantifies a 54.71% positive-delay rate in the cleaned data, identifies regional and shipping-mode bottlenecks, and provides a structured foundation for proactive late-delivery prediction.

From a recruiter perspective, the strongest capabilities demonstrated are Python/Pandas data preparation, KPI development, exploratory analysis, operational root-cause thinking, business visualization and leakage-aware machine-learning workflow design.

21. Project Summary

Supply Chain Delivery Performance & Delay Risk Analysis

One-line description: Analyzed 180K+ supply-chain records to quantify delivery delays, identify operational bottlenecks, assess profitability and design a predictive late-delivery risk workflow.

- Cleaned and transformed 180,519 raw records into a 172,765-row analytical dataset using Python/Pandas.
- Engineered delivery-delay, processing-time, calendar and profitability features and identified a 54.71% positive-delay rate.
- Performed bottleneck and regional driver analysis across shipping mode, region, segment, department, order status and transaction type.

- Built 12 business-focused visualizations to communicate operational performance and improvement priorities.
- Implemented Logistic Regression, Random Forest and SMOTE-based Balanced Random Forest workflows with leakage-aware feature selection.